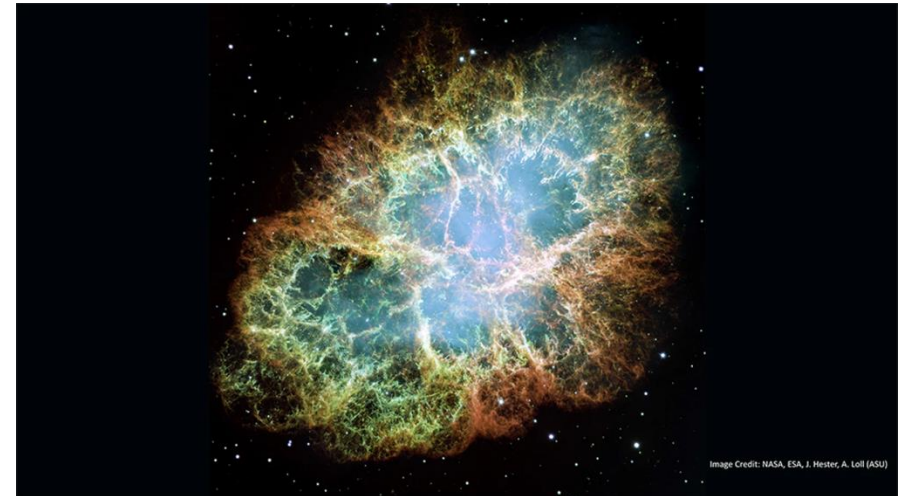


Rayleigh-Taylor Instability

Rayleigh-Taylor instability happens when a denser fluid is pushed through a lighter fluid.



Rayleigh-Taylor Instability: Geometry

Contact discontinuity between fluids 1 and 2, at rest.

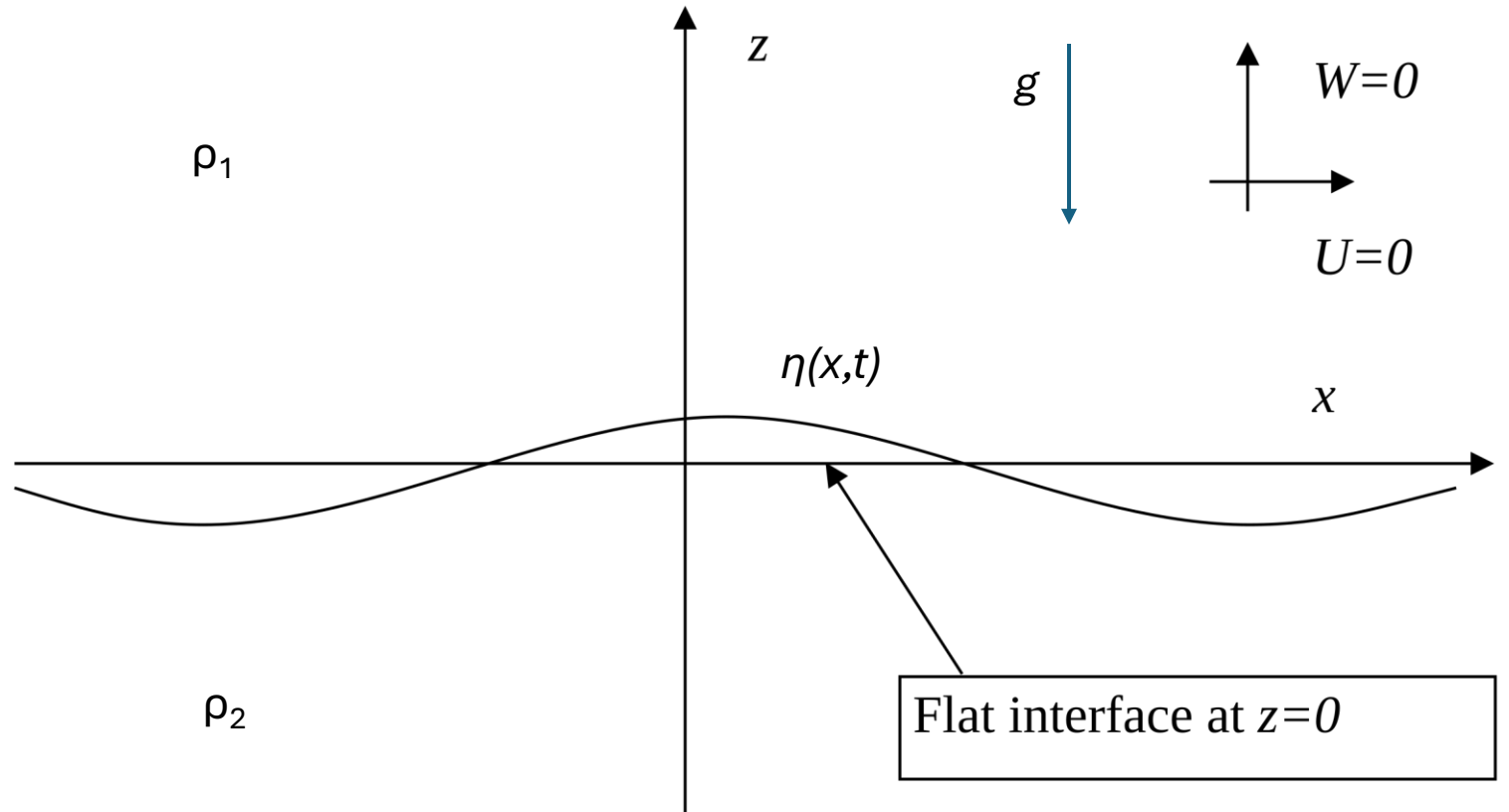
$\eta(x,t)$ is a perturbation of the interface.

The question we ask is: what is the oscillation frequency of $\eta(x,t)$?

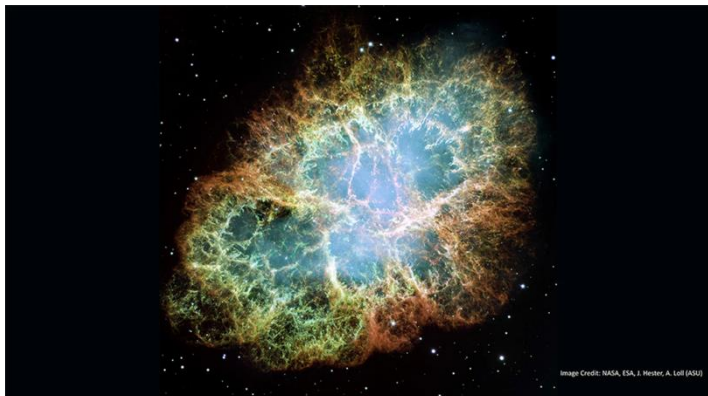
The solution is the dispersion relation

$$\omega^2 = \frac{(\rho_2 - \rho_1)}{(\rho_1 + \rho_2)} g k$$

Exponential growth (instability) exists when $\rho_1 > \rho_2$

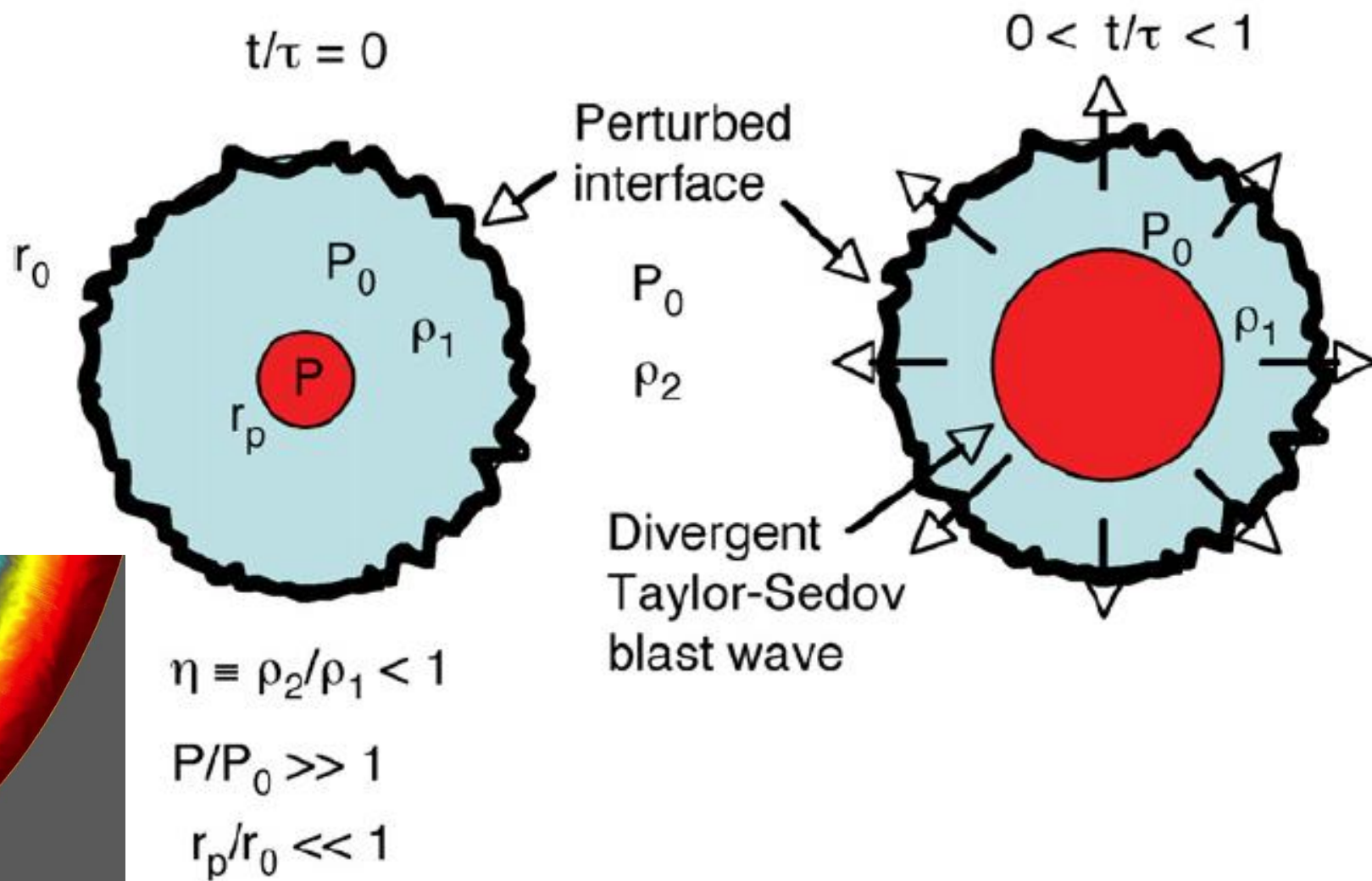
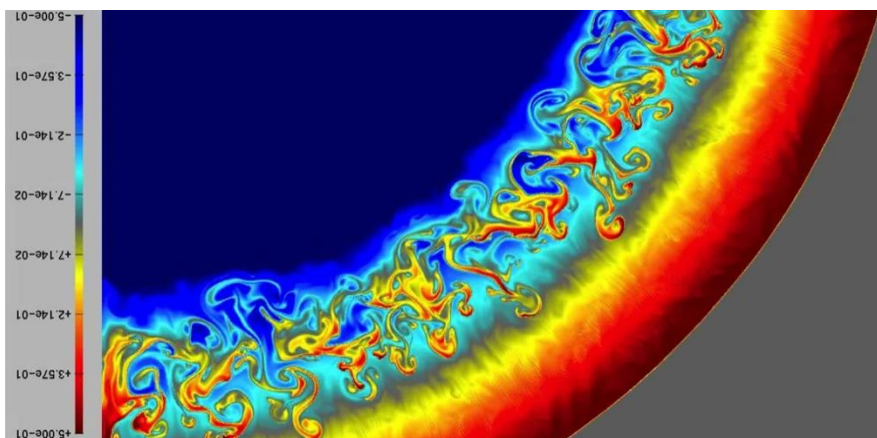


Rayleigh-Taylor Instability on Supernova Remnant



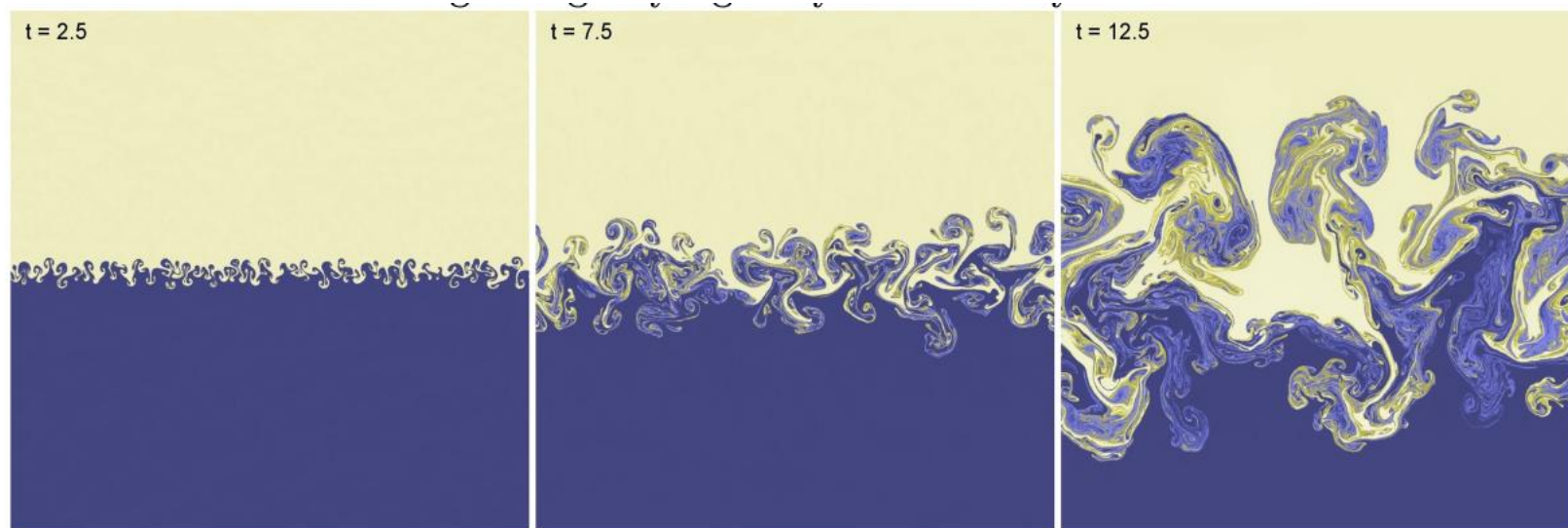
Heavy ejecta pushed through lighter ISM.

Rayleigh-Taylor unstable.



Dispersion relation

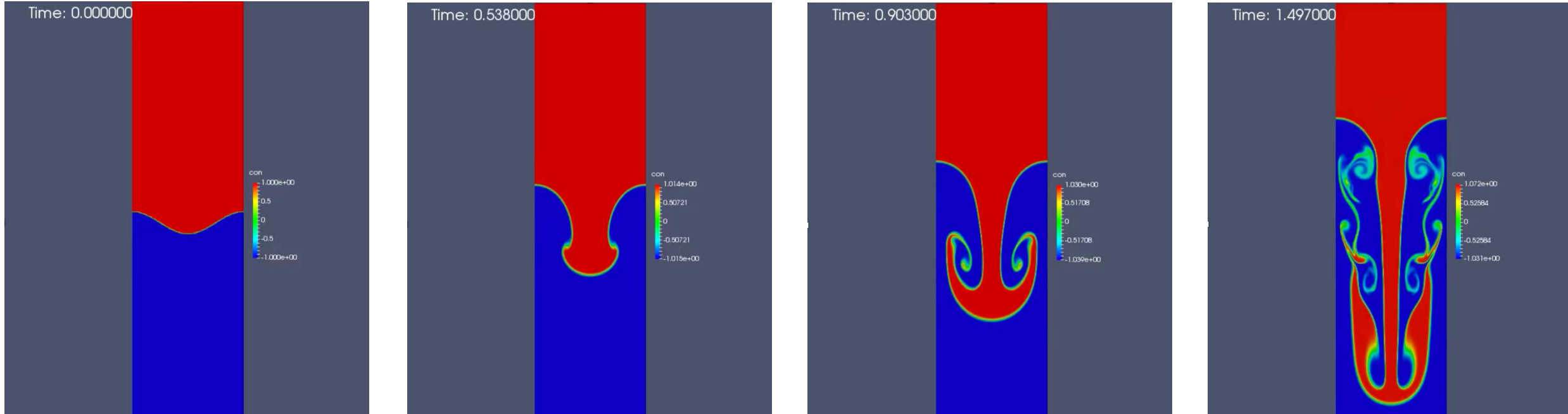
$$\omega^2 = \frac{(\rho_2 - \rho_1)}{(\rho_1 + \rho_2)} gk$$



Notice that the fastest growth occurs for the smallest scales.

Indeed, it is seen that the small scales grow first, then the larger.

Evolution of a single “finger”



A secondary instability lead to roll-up of the finger.

That's Kelvin-Helmholtz instability kicking in once shear develops (next class).